IDS Fundamentals

Intrusion detection systems are for if an attacker successfully bypasses a fire wall and is a security implementation that will detect malicious activity within a network in a timely manner.

Types of IDS

Deployment Modes

IDS can be deployed in the following ways:

- **Host Intrusion Detection System (HIDS):** Host-based IDS solutions are installed individually on the hosts and are responsible for only detecting potential security threats associated with that particular host. They provide detailed visibility of the host's activities. However, host intrusion detection systems can be challenging to manage in large networks as they are resource-intensive and require management on each host.
- **Network Intrusion Detection System (NIDS):** Network-based IDS solutions are crucial in detecting potentially malicious activities within the whole network, regardless of any specific hosts. They monitor the network traffic of all the hosts involved to detect suspicious activities. It provides a centralized view of all the detections inside the whole network.

Detection Modes

Signature-Based IDS: Many attacks occur every day. Each attack has its unique pattern, which is known as a signature. These signatures are preserved by the IDS in their databases so that if the same attack happens in the future, it gets detected by its signature and reported to the security administrators for action. Adding to the database proves for a more effective IDS

they cannot detect zero day exploits as zero day exploits have no previous signatures/patterns

Anomaly-Based IDS: This type of IDS first learns the normal behaviour (baseline) of the network or system and performs detections if there is any deviation from the normal behaviour. Anomaly-based IDS can also detect zero-day attacks because they don't rely on available signatures for detection

this type of IDS may generate a lot of false positives because the nature of most legitimate programs matches the malicious ones.

Hybrid IDS: A hybrid IDS combines the detection methods of signature-based IDS and anomaly-based IDS to leverage the strengths of each approach. Some known threats may already have some signatures in the IDS database; in this case, the hybrid IDS would use the

detection technique of the signature-based IDS. If it encounters a new threat, it can leverage the detection method of anomaly-based IDS

IDS Example: Snort

Snort is one of the most widely used open-source IDS solutions developed in 1998. It uses signature-based and anomaly-based detections to identify known threats

Snort's built-in rules can detect a lot of malicious traffic for you

can configure Snort to detect specific types of network traffic not covered by the default rule files

Modes of Snort

Mode	Description	Use Case
Packet sniffer mode	This mode reads and displays network packets without performing any analysis on them. The packet sniffer mode of Snort does not directly relate to IDS capabilities, but it can be helpful in network monitoring and troubleshooting. In some cases, system administrators might need to read the traffic flow without performing any detection to diagnose specific issues. In this case, they can utilize the packet sniffer mode of Snort. This mode allows you to display the network traffic on the console or even output it in a file.	The network team observes some network performance issues. To diagnose the issue, they need detailed insights into the network traffic. For this purpose, they can utilize Snort's packet sniffer mode.
Packet logging mode	Snort performs detection on the network traffic in real-time and displays the detections as alerts on the console for the security administrators to take action. However, in some cases, the network traffic needs to be logged for later analysis. The packet logging mode of Snort allows you to log the network traffic as a PCAP (standard packet capture format) file. This includes all the network traffic and any detections from it. Forensic investigators can use these Snort log files to perform the root cause analysis of previous attacks.	The security team needs to initiate a forensic investigation of a network attack. They would need the traffic logs to perform the root cause analysis. The network traffic logged through Snort's packet logging mode can help them.
Network Intrusion	Snort's NIDS mode is the primary mode that monitors network traffic in real-time and	The security team must proactively monitor their

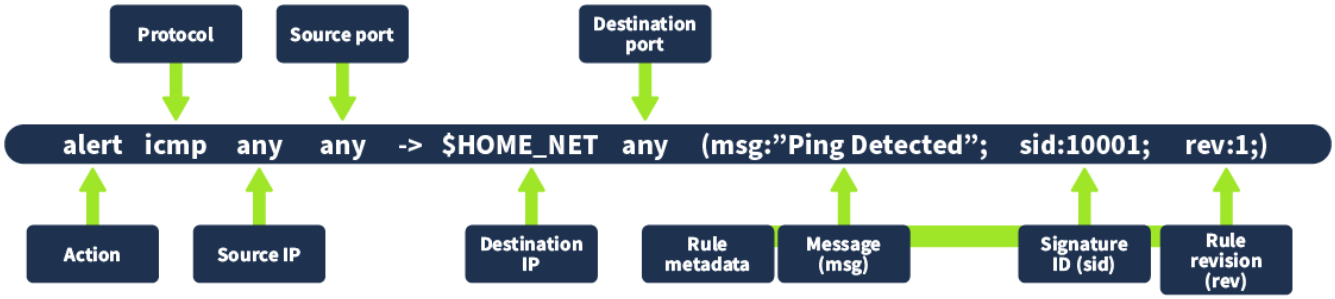
Mode	Description	Use Case
Detection System mode	applies its rule files to identify any match to the known attack patterns stored as signatures. If there is a match, it generates an alert. This mode provides the main functionality of an IDS solution.	network or systems to detect potential threats. They can leverage Snort's NIDS mode to achieve this.

Snort Usage

list all the files and folders present in Snort's main directory:

```
ubuntu@tryhackme:~$ ls /etc/snort
classification.config  reference.config  snort.debian.conf
community-sid-msg.map  rules             threshold.conf
gen-msg.map            snort.conf        unicode.map
```

Rule Format



The details of the components involved in this rule are given below:

- **Action:** This specifies which action to take when the rule triggers. In this case, we have the action to "alert" when the traffic matches this rule.
- **Protocol:** This refers to the protocol that matches this rule. In this case, we use the protocol "ICMP" when pinging a host.
- **Source IP:** This determines the IP originating from the traffic. Since we want to detect traffic from any source IP, we set this as "any".
- **Source port:** This determines the port from which the traffic originates. Since we want to detect traffic from any source port, we set this as "any".
- **Destination IP:** This specifies the destination IP to which the matching traffic comes; it generates the alert. In this case, we used "$HOME_NET". This is a variable, and we defined its value as our whole network's range in Snort's configuration file.
- **Destination port:** This specifies the port the traffic would reach. Since we want to detect traffic coming to any port, we set it to "any."
- **Rule metadata:** Every rule has some metadata. That is defined at the end of the rule in parentheses. The following are its components:
  - **Message (msg):** This describes the message displayed when the subject rule triggers. The message should indicate the type of activity detected. In this case, we used "Ping Detected".
  - **Signature ID (sid):** Every rule has a unique identifier that differentiates it from others. This identifier is called the signature ID (sid). In this case, we set the sid to "10001".
  - **Rule revision (rev):** This sets the rule's revision number. Every time the rule is modified, its revision number is incremented, which helps in tracking changes to any rule.

Rule Creation

Firstly, open the "local.rules" file in a text editor:

Edit Custom Rule File

```
ubuntu@tryhackme:~$ sudo nano /etc/snort/rules/local.rules
```

Now, add the following rule after the already present rules to the file:

```
alert icmp any any -> 127.0.0.1 any (msg:"Loopback Ping Detected"; sid:10003; rev:1;)
```

Rule Testing

start the Snort tool to detect any intrusions defined in the rule file. For this, we have to execute the following command with sudo privileges in our console:

```
sudo snort -q -l /var/log/snort -i lo -A console -c /etc/snort/snort.conf
```

Running Snort for Detections

```
ubuntu@tryhackme:~$ sudo snort -q -l /var/log/snort -i lo -A console -c /etc/snort/snor
07/24-10:46:52.401504  [**] [1:1000001:1] Loopback Ping Detected [**] [Priority: 0] {IC
07/24-10:46:53.406552  [**] [1:1000001:1] Loopback Ping Detected [**] [Priority: 0] {IC
07/24-10:46:54.410544  [**] [1:1000001:1] Loopback Ping Detected [**] [Priority: 0] {IC
```

Running Snort on PCAP Files

historical network traffic logged in a file, and you have to perform a forensic investigation to determine any signs of intrusion through that traffic. This traffic is usually logged in the standard packet capture format "PCAP". Snort is also equipped to perform detections on these PCAP files containing historical network traffic.

The following command with sudo privilege can be used to perform this action:

```
sudo snort -q -l /var/log/snort -r Task.pcap -A console -c /etc/snort/snort.conf
```

Practical Lab

run snort on Intro_to_IDS.pcap file

```
ubuntu@tryhackme:/etc/snort$ sudo snort -q -l /var/log/snort -r Intro_to_IDS.pcap -A console -c /etc/snort/snort.c
onf
07/18-12:52:59.337559  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:52:59.510662  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:52:59.510663  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:52:59.693619  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:52:59.696547  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:52:59.696584  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:52:59.867507  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:52:59.868435  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:53:00.050645  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:53:00.057313  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:53:00.272189  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:53:00.444487  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:53:00.444488  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:53:00.623967  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:53:00.647745  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
07/18-12:53:00.830895  [**] [1:1000002:1] SSH Connection Detected [**] [Priority: 0] {TCP} 10.11.90.211:54334 -> 1
0.10.161.151:22
```

Here i found the Ip address of the machine that tried to connect to the target machine using ssh : 10.11.90.211

```
0.10.161.151:22
07/18-12:53:16.954348  [**] [1:1000001:1] Ping Detected [**] [Priority: 0] {ICMP} 10.11.90.211 -> 10.10.161.151
07/18-12:53:17.956812  [**] [1:1000001:1] Ping Detected [**] [Priority: 0] {ICMP} 10.11.90.211 -> 10.10.161.151
07/18-12:53:18.972925  [**] [1:1000001:1] Ping Detected [**] [Priority: 0] {ICMP} 10.11.90.211 -> 10.10.161.151
ubuntu@tryhackme:/etc/snort$
```

there is another rule message that is picked up in the pcap file which is ping detected

the sid of the rule that detects ssh was 1000002